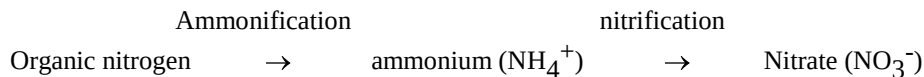


Nitrogen Transformation in Compost

How does the Nitrogen change through the composting process?

Nitrogen exists in both organic and inorganic forms in organic matter. Organic nitrogen consists of compounds such as proteins, urea, nucleic acids, and microbial biomass. Microorganisms mineralizes organic nitrogen to produce inorganic forms of nitrogen such as: ammonia, nitrite, and nitrate. The inorganic forms of nitrogen are available as nutrients for plants.

When composting conditions are aerobic, most of the nitrogen is converted from ammonia to nitrate. This is called nitrification. Both ammonia and nitrate can accumulate in composts if other chemical conditions allow for it.



When the pH of the compost is within the range of 5 - 8, most of the ammonia accumulates in the compost in the form of ammonium and since there are high levels of sulphur compounds in compost, most of the ammonium is thought to exist in the form of ammonium sulphate ((NH₄)₂SO₄) under aerobic conditions.

Where has all the nitrogen gone?

The amount of total nitrogen (organic and inorganic) that is in the finished compost is usually lower then the raw manure. This could be due to several reasons. One reason is simply due to the rate of carbon amendment to the compost. In order to compost optimally, a Carbon to Nitrogen ratio (C:N ratio) of about 30 - 35:1 should be attained. By adding carbon materials to the manure will result in a dilution of the amount of nitrogen in the final mix.

The production of ammonia is caused by high levels of nitrogen, which is indicated by a low C:N ratio (e.g. 16:1). Increasing the C/N ratio will correct this symptom and reduce the amount of nitrogen loss as ammonia. A high pH (pH 9 - 12) may also cause ammonia release since ammonium ions start to become unstable in the compost and is released as ammonia.

Composting process optimization is also important in preventing the loss of nitrogen to the atmosphere as ammonia. Under anaerobic conditions there are large amounts of ammonia that is produced from ammonification of organic compounds which cannot be oxidized by nitrification processes to form nitrates. Therefore, to conserve nitrogen in composts it is crucial to maintain aerobic conditions. Apart from ammonia, nitrogen in compost is also loss as nitrogen gas (N₂) and Nitrous oxides (NO and N₂O). Nitrogen gas does not pose an environmental concern, however nitrous oxides are considered greenhouse gases. Nitrous oxides are also a result of anaerobic processes and under optimal conditions these compounds should be converted to nitrogen gas (N₂) by microorganisms.

When compost piles are not managed effectively much of the nitrogen can be loss in the leachate. Nitrates in the leachate can pose environmental, and health and safety issues with ground water pollution.

What are some typical levels of nitrogen in finished compost?

Compost	Final Nitrogen levels (%)
Feedlot Cattle manure with bedding	1.35
Poultry litter / wood chips	1.80
Hog Manure / wood chips	2.00
Municipal Solid Waste	1.20
Yard Waste	0.76
Pulp Sludge	1.30

Results are from trials conducted at the Olds College Composting Technology Centre.

In summary, nitrogen conservation in composts starts and ends with effective composting process management. There will naturally be some loss of nitrogen to the atmosphere. However, the losses can be controlled and minimized by process optimization. The compost pile needs to be aerobic and moisture and pH levels should be closely monitored throughout the composting process.